Type MS · Technology, the environment and health

Answer 1 of 4 · Question 5 part c · 6 marks

2025 JUN · Paper 01 · extract 1 of 5 · find part c in this mark scheme extract

5(a)(i)	Award **one** mark for any of the following up to a maximum of **two** marks: • Heart rate (1) to measure blood flow/pulse rate (1) • Barometric pressure (1) to determine altitude (1) • Microphone (1) for voice activation control (1) • Temperature/Thermometer (1) to measure ambient/body temperature (1) • GPS (1) to calculate longitude/latitude (1) • Light (1) to adjust screen brightness/charge watch battery (1) • Magnetometer (1) to act as a compass (1) • ECG (1) measure heart rhythm (1) • Blood pressure monitor (1) measure pressure in arteries / can report heart rate/pulse rate (1)	Allow any reasonable sensor that might be embedded within a smart watch. Purpose must be linked to the sensor Purpose must expand on name (e.g. heart rate sensor - monitor user's heart rate is not sufficient for second mark) Allow purpose mentioned in second column if nothing in first column	**2**
Question	Answer	Additional Guidance	Mark
5(a)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It has a low power requirement (1) • It can use a low processor speed (1) • It has built in/integrated storage (1) • It relies on flash memory (1) • It can be mass produced at a low cost per device (1) • It is small (1) • It can work in a large range of operating temperature ranges (1) • It is robust (1) • It has its operating software set by the manufacturer (1) • It is designed to do one particular task (1)		**2**

Type MS · Technology, the environment and health

Answer 1 of 4 · Question 5 part c · 6 marks

2025 JUN · Paper 01 · extract 2 of 5 · find part c in this mark scheme extract

Question	Answer	Additional Guidance	Mark
5(b)(i)	**The only correct answer is B** *A is not correct because low-level languages are not converted into high-level languages* *C is not correct because assemblers translate assembly code to produce machine code but do not execute it* *D is not correct because compilers and interpreters translate high-level languages*		1
Question	Answer	Additional Guidance	Mark
5(b)(ii)	Award up to **two** marks for a linked explanation such as: • Makes best use of the hardware (1) because they can directly access (specific) hardware (features) (1) • It reduces manufacturing costs (1) because of a lower processor specification/expense (1) • Optimise their code for hardware (1) directly access/control registers (1)	Do not accept references to translation time	2
Question	Answer	Additional Guidance	Mark
5(b)(iii)	Award **one** mark for any of the following up to a maximum of **two** marks: • Abide by the law / avoid plagiarism / copyright infringements (1) • Adhere to a professional code of conduct (1) • Maintain/enhance their skills through further learning/keep up to date/attend (relevant) conferences (1) • Test code thoroughly/do not release code with known errors/vulnerabilities (1) • Respect privacy and security of data (1) • Preserve the environment (1) • Make software accessible/unbiased (1)		2

Type MS · Technology, the environment and health

Answer 1 of 4 · Question 5 part c · 6 marks

2025 JUN · Paper 01 · extract 3 of 5 · find part c in this mark scheme extract

Question	Answer	Additional Guidance	Mark
5(b)(iv)	**The only correct answer is A** *B is not correct because 500 KiB is 512,000 bytes* *C is not correct because 0.5 MiB is 524,288 bytes* *D is not correct because 5MB is 5,000,000 bytes*		**1**
Question	**Answer**	**Additional Guidance**	**Mark**
5(c)	Indicative content: • Can use sensors to take/track readings such as heart rate • Can track location via GPS and record routes taken/distance travelled • Can calculate exercise/movement undertaken such as steps/calories burned • Can be linked via 5G/Internet Of Things/Personal Area Network (PAN) to other devices/apps that can record and analyse the data • Can trigger alerts when users are inactive • Can monitor blood oxygen/blood sugar/sleep patterns • Can applaud/give kudos when an achievement/event takes place • Can increase motivation of the wearer to be healthy/perform exercise due to feedback loop • Workouts/alerts can be synced to the watch • Linked apps can perform more in-depth analysis • No guarantee that users will buy into benefits and lead a healthier lifestyle • Monitoring vulnerable people allowing independent living	Allow content referring to other wearable technology which is not a smartwatch	**6**
		Total for question 5	**16**

Type MS · Technology, the environment and health

Answer 1 of 4 · Question 5 part c · 6 marks

2025 JUN · Paper 01 · extract 4 of 5 · find part c in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1-2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

16

Type MS · Technology, the environment and health

Answer 1 of 4 · Question 5 part c · 6 marks

2025 JUN · Paper 01 · extract 5 of 5 · find part c in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Technology, the environment and health

Answer 2 of 4 · Question 6 part c · 6 marks

2021 JUN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

6(a)(i)	Award **two** marks for a linked explanation such as: • Uses DNA/biomolecular components (1) rather that standard hardware/silicon chips (1) • DNA uses four-character genetic alphabet (1) rather than binary digits (1) • DNA uses chemical reactions (1) rather than electrical states/properties (1) • DNA computing uses massive parallel processing/strands (1) to solve problems that otherwise would take impossible amounts of time (1)		2
6(a)(ii)	Award **one** mark from: • The ability of a system to be in multiple states at the same time (1) • Multiple probabilities at the same time (1) • One state combines all possibilities (1) • The state is not known until it is measured (1)		1

Type MS · Technology, the environment and health

Answer 2 of 4 · Question 6 part c · 6 marks

2021 JUN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **two** marks from: • Some people may not want to use technology (1) • Some people may not be able to afford to use technology (1) • Some people may not have access to the technology/infrastructure (1) • Some people may not be allowed access to technology (1) • Some people may not be able to use it (disabilities etc.) (1)		2

Question Number	Answer	Additional Guidance	Mark
6(c)	Indicative content Impacts • Pollution (water, air, noise) resulting from the manufacturing process • High energy and water volumes needed for the manufacturing process • Mineral mining contaminates ground water • Mineral (copper, gold, silver, lithium) resource depletion • High energy use to keep machines cool with air conditioning or fans • Contain toxic components which means computing devices are hazardous waste • Batteries (laptop, lithium cells) disposal is hazardous and needs specialised disposal facilities • Computing devices sent to landfills contaminate ground and water resources Actions • Check national and local legislation • Some countries/states/regions require sellers to recycle own waste • Research recycling facilities to see if e-waste is acceptable • Find if local/national/regional government agencies charge for collection and disposal of e-waste • Find sellers offering exchange old for new + cost purchase options • Find charities (local, regional) which take donations of old devices • Turn off machines when not in use to save energy		6

Type MS · Technology, the environment and health

Answer 2 of 4 · Question 6 part c · 6 marks

2021 JUN · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content
Level 1	1-2	Basic independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent and logically structured.

Type MS · Technology, the environment and health

Answer 2 of 4 · Question 6 part c · 6 marks

2021 JUN · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

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Type MS · Technology, the environment and health

Answer 3 of 4 · Question 3 part e.i · 1 mark

2021 NOV · Paper 01 · extract 1 of 5 · find part e.i in this mark scheme extract

3(a)	Award **one** mark for correct network type Award **one** mark for correct usage model One mark for each column		2

Network type	Tick (□)	Usage model	Tick (□)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	□
Personal Area Network (PAN)	□		

Type MS · Technology, the environment and health

Answer 3 of 4 · Question 3 part e.i · 1 mark

2021 NOV · Paper 01 · extract 2 of 5 · find part e.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)</td><td>Award one mark for each correct layer name up to a maximum of two
Award one mark per layer for a correct function associated with the correct layer up to a maximum of two

<table><tr><th>Layer</th><th>Function</th></tr><tr><td>Application</td><td>• Selects and uses the correct protocol to transmit data
• Interacts with the user</td></tr><tr><td>Transport</td><td>• Identifies the server port to use
• Identifies the client port to use
• Divides the data into packets
• Numbers the packets
• Adds the total number of packets
• Sets up communication between hosts / Establishes end to end communication
• Passes the packets to the network layer
• Checks the packets arrive at the destination
• Resends any packets that have not arrived</td></tr><tr><td>Network</td><td>• Adds the source/sender's IP address
• Adds the destination IP address
• Routes the packets</td></tr><tr><td>Data link</td><td>• Controls physical connections between pieces of hardware
• Adds MAC addresses to the packets
• Sends the packets on their way
• Adds headers and trailers</td></tr></table></td><td>Accept functions at the receiving end too</td><td>4</td></tr>
</table>

Type MS · Technology, the environment and health

Answer 3 of 4 · Question 3 part e.i · 1 mark

2021 NOV · Paper 01 · extract 3 of 5 · find part e.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	**The only correct answer is A** B *is not correct because* it is not a smartphone frequency band C *is not correct because* it is not a smartphone frequency band D *is not correct because* it is not a smartphone frequency band		1

Question Number	Answer	Additional Guidance	Mark
3(d)(i)	Award **one** mark from: • higher frequency has more waves per second/carries more data in the same time (1) • higher frequency gives greater bandwidth (1) • higher frequency is more stable/less prone to interference (1)		1
3(d)(ii)	Award up to **two** marks for a linked explanation such as: • gives Carlo faster responses to his communications/connections (1) allowing Carlo to move/download/share data in a shorter time (1) • gives Carlo more reliable communications/can access communications more widely/in more places (1) giving Carlo a better user experience (1) • gives Carlo more secure communications (1) improving his privacy/reducing other people's ability to intercept/spy on his communications (1)		2

Type MS · Technology, the environment and health

Answer 3 of 4 · Question 3 part e.i · 1 mark

2021 NOV · Paper 01 · extract 4 of 5 · find part e.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(i)	Award **one** mark from: • do not leave the smartphone on standby/turn off when not being used (1) • use (portable) solar cells/other renewable generation method (1) • unplug charger when not in use (1) • reduce screen brightness (1) • turn off WiFi/GPS/Bluetooth/location (when not in use) (1) • close apps that are not being used (1)		1

Question Number	Answer	Additional Guidance	Mark
3(e)(ii)	Award up to **two** marks for a linked explanation. Award **one** mark for the action • keep smartphone for longer (1) • repair/recycle smartphone if not working (1) • upgrade the operating system (1) Award **one** mark for the impact • less e-waste is generated (1) • manufacturing could be reduced (1)		2
		Total	**13**

Type MS · Technology, the environment and health

Answer 3 of 4 · Question 3 part e.i · 1 mark

2021 NOV · Paper 01 · extract 5 of 5 · find part e.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Technology, the environment and health

Answer 4 of 4 · Question 3 part e.ii · 2 marks

2021 NOV · Paper 01 · extract 1 of 5 · find part e.ii in this mark scheme extract

3(a)	Award **one** mark for correct network type Award **one** mark for correct usage model One mark for each column		2

Network type	Tick (□)	Usage model	Tick (□)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	□
Personal Area Network (PAN)	□		

Type MS · Technology, the environment and health

Answer 4 of 4 · Question 3 part e.ii · 2 marks

2021 NOV · Paper 01 · extract 2 of 5 · find part e.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)</td><td>Award one mark for each correct layer name up to a maximum of two
Award one mark per layer for a correct function associated with the correct layer up to a maximum of two

<table><tr><th>Layer</th><th>Function</th></tr><tr><td>Application</td><td>• Selects and uses the correct protocol to transmit data
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Type MS · Technology, the environment and health

Answer 4 of 4 · Question 3 part e.ii · 2 marks

2021 NOV · Paper 01 · extract 3 of 5 · find part e.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	**The only correct answer is A** B *is not correct because* it is not a smartphone frequency band C *is not correct because* it is not a smartphone frequency band D *is not correct because* it is not a smartphone frequency band		1

Question Number	Answer	Additional Guidance	Mark
3(d)(i)	Award **one** mark from: • higher frequency has more waves per second/carries more data in the same time (1) • higher frequency gives greater bandwidth (1) • higher frequency is more stable/less prone to interference (1)		1
3(d)(ii)	Award up to **two** marks for a linked explanation such as: • gives Carlo faster responses to his communications/connections (1) allowing Carlo to move/download/share data in a shorter time (1) • gives Carlo more reliable communications/can access communications more widely/in more places (1) giving Carlo a better user experience (1) • gives Carlo more secure communications (1) improving his privacy/reducing other people's ability to intercept/spy on his communications (1)		2

Type MS · Technology, the environment and health

Answer 4 of 4 · Question 3 part e.ii · 2 marks

2021 NOV · Paper 01 · extract 4 of 5 · find part e.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(i)	Award **one** mark from: • do not leave the smartphone on standby/turn off when not being used (1) • use (portable) solar cells/other renewable generation method (1) • unplug charger when not in use (1) • reduce screen brightness (1) • turn off WiFi/GPS/Bluetooth/location (when not in use) (1) • close apps that are not being used (1)		1

Question Number	Answer	Additional Guidance	Mark
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		Total	**13**

Type MS · Technology, the environment and health

Answer 4 of 4 · Question 3 part e.ii · 2 marks

2021 NOV · Paper 01 · extract 5 of 5 · find part e.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark